

## 7. Functions and Methods



### Defining and calling functions in Python.

- Function is a block of code that we can use it again and again.

#### ➤ Types

##### 1. Built in function.

- Print()
- Input()

##### 2. User define function.

- Function without parameters without return type
- function with parameters without return type
- function without parameters with return type
- function with parameters with return type

#### ➤ Defining a Function

- You define a function using the def keyword, followed by the function name and parameters.

#### ➤ Calling a Function

- To call a function, use its name followed by parentheses, and pass the required arguments.

## ➤ Basic Syntax

- `function_name(parameters):`
- `"""docstring (optional)"""`
- `# function body`
- `return value # optional`

## ➤ Calling Functions

- `greet("Alice")` # Output: Hello, Alice!



Function arguments (positional, keyword, default).

## ➤ What is an Argument?

- In Python, an argument is a value you pass to a function when calling it.

### 1. Positional Arguments

- These are the most common type — values are assigned to parameters based on their position in the function call.

### 2. Keyword Arguments

- You specify arguments using parameter names, so the order doesn't matter.

### 3. Default Arguments

- You can assign default values to parameters. If a value is not provided during the call, the default is used.



## Scope of variables in Python.

- The scope of a variable refers to the part of the program where that variable is accessible or visible.

### ➤ Types of Variable Scope

#### 1. Local Scope

- Variables defined inside a function are only accessible within that function.

#### 2. Global Scope

- Variables defined outside all functions are global and accessible anywhere in the file.



## Built-in methods for strings, lists, etc.

- Python provides a rich set of built-in methods that can be used with various data types like strings, lists, tuples, and more.

## ➤ String Methods(for str type)

- Purpose: To manipulate and analyze text data.
- Examples of common methods:
- `lower()` – Converts all characters to lowercase.
- `upper()` – Converts all characters to uppercase.
- `replace()` – Replaces parts of the string.
- `find()` – Finds the position of a substring.

## ➤ List Methods (for list type)

- Purpose: To add, remove, sort, or find elements in a list.
- Examples of common methods:
- `insert()` – Inserts an item at a specific position.
- `remove()` – Removes the first matching item.
- `count()` – Counts how many times a value appears.